Muhammad Shaheer Khan

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Summary

Computer Engineering student at Middle East Technical University (METU) with industry internship experience in machine learning and computer vision. Hands-on experience in feature engineering, imbalanced classification, YOLOv8 object detection, and model export using OpenVINO and TensorRT. Seeking opportunities in applied machine learning and computer vision.

Education

Middle East Technical University (METU) Ankara, Turkey
B.Sc. in Computer Engineering *Sep 2022 – Present*

- **Current Coursework:** Image Processing, Introduction to Robotics, Introduction to Computer Graphics, Software Construction, Senior Design Project.

Experience

Machine Learning Intern Summer 2026
Optiway Solutions *Ankara, Turkey*

- Curated a transformer dissolved gas analysis (DGA) dataset by removing duplicates and correcting mislabeled and misaligned records, retaining 2,387 of 2,825 samples for model development.
- Engineered domain-specific gas-ratio features for seven-class fault classification, improving XGBoost accuracy from 81.5% to 90.4% and Random Forest accuracy from 81.9% to 89.7%.
- Compared SMOTE, Borderline-SMOTE, ADASYN, CTGAN, and TVAE for minority-class oversampling, evaluating macro F1 and class-wise performance.
- Tuned Random Forest and XGBoost with GridSearchCV and RandomizedSearchCV using stratified five-fold cross-validation; evaluated generalization on held-out test data.

Computer Vision Intern Summer 2025
Aremak *Ankara, Turkey*

- Trained YOLOv8 models on a custom dataset for real-time people detection, achieving 0.91 mAP and inference speeds above 35 FPS.
- Compared YOLOv8 variants using precision, recall, mAP, and confusion matrices to assess detection performance and inference-speed trade-offs.
- Exported trained models to OpenVINO and TensorRT for deployment on edge devices.

Projects

HOOPS – Basketball Video Analysis *Ongoing senior project*

- Planning an offline system for recorded basketball footage, with player/ball detection, tracking, court mapping and analytics as planned features.
- Prepared the development plan covering delivery stages, integration, team coordination and evaluation.

Technical Skills

Programming: Python, C++, C
Machine Learning: scikit-learn, XGBoost, Random Forest, imbalanced-learn, SDV (CTGAN, TVAE)
Computer Vision: YOLOv8, Object Detection, Dataset Preparation, Model Evaluation
Data Analysis: NumPy, pandas
Model Deployment: OpenVINO, TensorRT
Tools: Git, Jupyter Notebook, Google Colab, VS Code
Operating Systems: Linux (Basic)

Languages

English – Fluent | **Urdu** – Native | **Turkish** – A2